RECOMMENDATION

UnitedHealth Group (UNH): Long

Base-case DCF value **$526.64** per share against an as-of price of **$367.08**: **+43.5%**.

$354.47	$526.64	$764.68
bear (-3.4%)	base (+43.5%)	bull (+108.3%)

As of **2026-09-30**. The value is an intrinsic-value estimate, not a dated price target. The bear case sits *below* the price, and the result depends on a low beta (slide 11). Educational project, not investment advice.

WHY

Thesis in three bullets

1. **The price assumes little growth.** The reverse DCF says the price implies **6.0%** stage-1 FCF growth. UNH's median rolling 5-year FCF growth is **13.7%**.
2. **Revenue kept growing.** Revenue grew from **$287.6B** (FY2021) to **$447.6B** (FY2025). The problem is margin, not demand.
3. **The thesis is a margin recovery, and it is not yet in the numbers.** Operating margin fell from **8.7%** to **4.2%**. The long case needs FCF to stop shrinking.

FUNDAMENTALS

Business snapshot

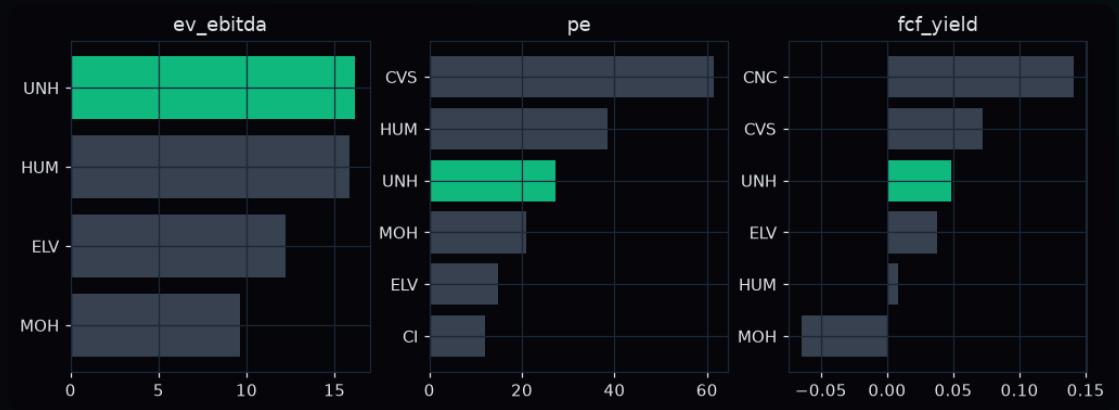
Fiscal year	Revenue	Operating income	Op. margin	FCF	FCF margin
FY2021	**$287.6B**	**$24.0B**	**8.3%**	**$19.9B**	**6.9%**
FY2022	**$324.2B**	**$28.4B**	**8.8%**	**$23.4B**	**7.2%**
FY2023	**$371.6B**	**$32.4B**	**8.7%**	**$25.7B**	**6.9%**
FY2024	**$400.3B**	**$32.3B**	**8.1%**	**$20.7B**	**5.2%**
FY2025	**$447.6B**	**$19.0B**	**4.2%**	**$16.1B**	**3.6%**

SEC EDGAR 10-K figures as first reported. FCF = operating cash flow − capex. Revenue keeps rising while operating income and FCF shrink. That gap is the key driver to watch.

RELATIVE VALUE

Comps: a premium on earnings, a discount on cash

	EV/EBITDA	P/E	FCF yield	Gaps
UNH	**16.2x**	**27.3x**	**4.9%**	
ELV	**12.2x**	**14.9x**	**3.8%**	
CI	n/a	**12.1x**	n/a	capex not reported in XBRL; total debt not reported in XBRL
HUM	**15.9x**	**38.6x**	**0.8%**	
CNC	n/a	n/a	**14.1%**	net loss: P/E not meaningful; negative EBITDA: EV/EBITDA not meaningful
MOH	**9.6x**	**20.9x**	**-6.4%**	
CVS	n/a	**61.5x**	**7.2%**	total debt not reported in XBRL
Peer median	**12.2x**	**20.9x**	**3.8%**	

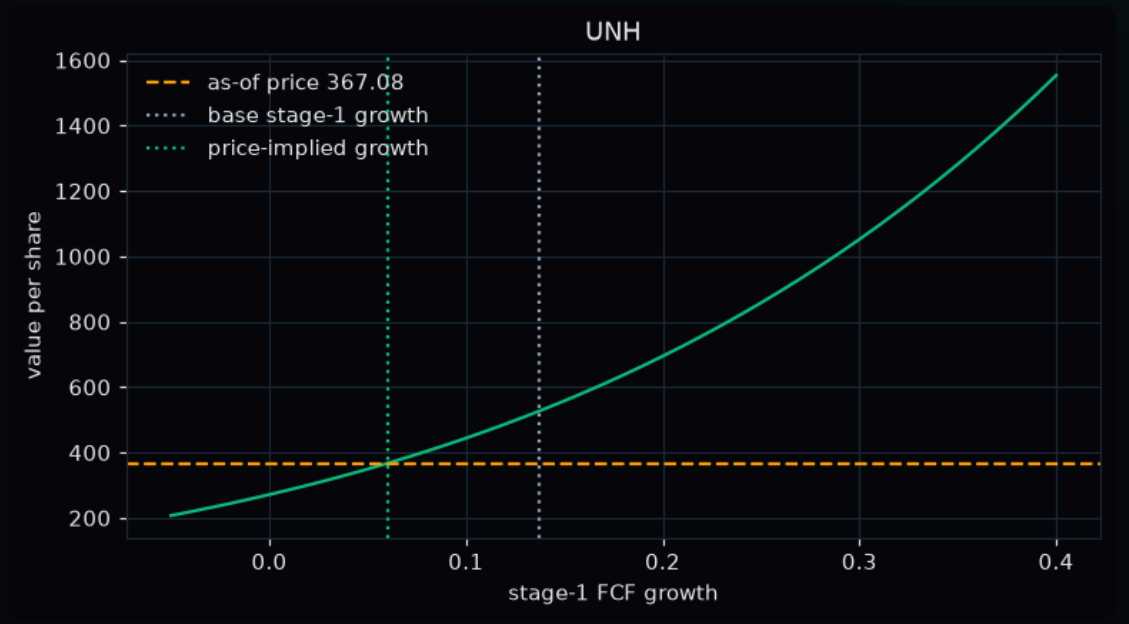


UNH trades above the peer median on P/E and EV/EBITDA but offers a higher FCF yield. Comps alone don't make the case. Gaps are left blank, never estimated.

VALUATION

DCF and reverse DCF: what the price assumes

risk free	**5.3%**	FRED DGS10 on 2026-09-30
beta	**0.41**	5y daily stock vs SPY returns to as-of (OLS slope)
equity risk premium	**8.3%**	Ken French Mkt-RF, arithmetic mean 1926-07 to 202608, x12
cost of equity	**8.7%**	CAPM: rf + beta x ERP
pre tax cost of debt	**5.5%**	FY2025 interest expense / average total debt (SEC 10-K)
tax rate	**12.9%**	FY2025 income tax / pretax income (SEC 10-K)
equity weight	**82.0%**	market cap (as-of price x Yahoo shares) / (market cap + book debt)
wacc	**8.0%**	we x ke + (1 - we) x kd x (1 - t)

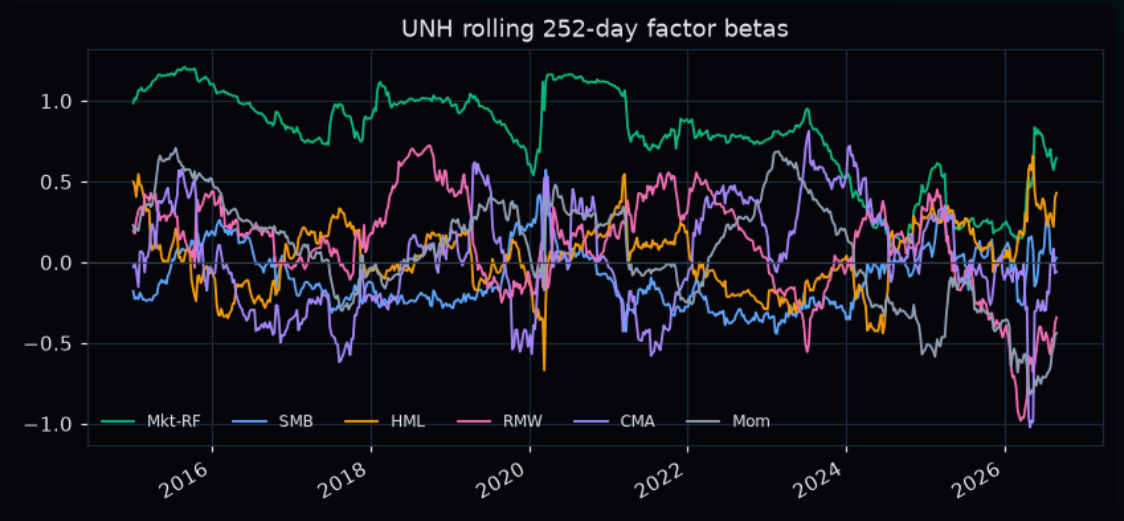


Base FCF **$16.1B** (FY2025 operating cash flow - capex (SEC 10-K)). Stage-1 growth **13.7%** fades to **3.0%** terminal growth over **10** years. Value **$526.64**. The price implies **6.0%** growth.

EXPOSURES

Factor exposures: what you are really buying

Factor	Beta	t-stat
Mkt-RF	+0.82	13.3
SMB	-0.12	-2.0
HML	+0.13	1.8
RMW	+0.26	4.3
CMA	+0.09	0.8
Mom	+0.08	1.7
Alpha (annual)	5.9%	0.8



FF5 plus momentum, daily, **2014-01-03** to **2026-08-31**. R^2 **0.23**: most of UNH's risk is stock-specific. After the market, the profitability (RMW) tilt has the strongest t-stat. Alpha is not significant.

RISK

Risk profile

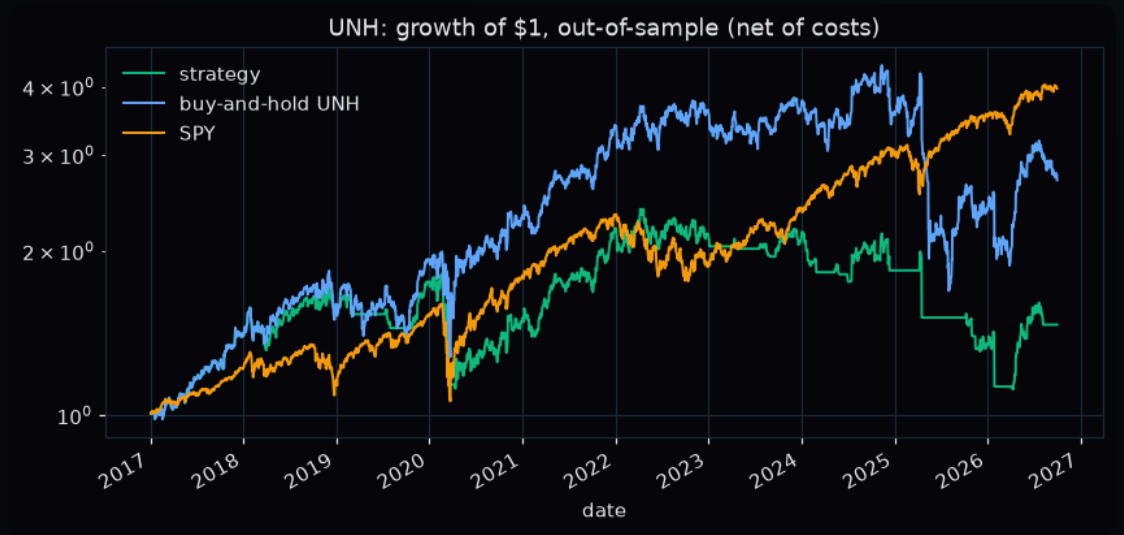
	Ann. vol	Beta vs SPY	Max drawdown	1-day 95% VaR	1-day 95% ES
Full sample: UNH	**28.6%**	**0.78**	**-61.4%**	**2.3%**	**4.0%**
Full sample: SPY	**17.1%**	**1.00**	**-33.7%**	**1.6%**	**2.6%**
Trailing 3y: UNH	**36.8%**	**0.25**	**-61.4%**	**2.8%**	**5.8%**
Trailing 3y: SPY	**15.3%**	**1.00**	**-18.8%**	**1.4%**	**2.1%**

UNH is more volatile than SPY and has suffered a **-61.4%** drawdown. Its low recent beta (**0.25** trailing 3y) reflects stock-specific moves, not safety.

EVIDENCE

Backtested signal: it does not help

	CAGR	Sharpe (rf = 0)	Max DD
Trend signal	4.0%	0.29	-53.3%
Buy-and-hold UNH	10.8%	0.49	-61.4%
SPY	15.2%	0.87	-33.7%



Out-of-sample **2017-01-03** to **2026-09-30**, walk-forward window choice, next-day execution, **10** bp per switch. Both the signal and buy-and-hold UNH trail SPY on Sharpe. This is a valuation call, not a momentum trade.

SCENARIOS

Bull, base and bear

	Stage-1 growth	WACC	Value	vs price
Bear	7.6%	8.5%	$354.47	-3.4%
Base	13.7%	8.0%	$526.64	+43.5%
Bull	19.4%	7.5%	$764.68	+108.3%



The bear case is -3.4%, below today's price. Growth is set at the lower quartile, median and upper quartile of UNH's own rolling 5-year FCF growth, and WACC moves by ±**0.5%**. No probability weights are applied, because there is no data-based way to set them. All three cases still use the 5y beta that sets WACC at **8.0%**. With the full-sample beta, the base case alone falls to **-12.5%** (slide 11).

CATALYSTS

What to monitor

- **Next 10-K FCF.** FY2025 FCF was **$16.1B**. The price implies **6.0%** a year from here. A further decline would break the thesis.
- **Operating margin.** It is **4.2%** against **8.7%** in FY2023. A recovery is the upside driver.
- **Beta re-estimate.** If the 5y beta drifts toward the full-sample **0.78**, WACC rises to **10.5%**.

No dated catalysts are listed: this project uses only the frozen data and does not forecast events.

RISKS

Key risks and what would change my mind

- **The bear case loses money.** The bear value of **$354.47** is **-3.4%** vs the price.
- **The discount rate rests on a low beta.** The 5y beta of **0.41** gives the **8.0%** WACC. The full-sample beta of **0.78** gives a **10.5%** WACC and a value of **$321.29** (**-12.5%**). *On that beta the long case disappears.*
- **FCF is falling.** It fell from **$25.7B** (FY2023) to **$16.1B** (FY2025), while stage-1 growth uses the **13.7%** long-run median.
- **The track record is weak.** Buy-and-hold UNH's Sharpe of **0.49** trails SPY's **0.87**, with a **-61.4%** max drawdown.

I would drop the long if the next 10-K shows FCF still falling, or if a re-estimated beta pushes WACC to the point where the base value falls below the price.

METHOD

Methodology and reproducibility

- As-of **2026-09-30**. Data is frozen with SHA-256 hashes: Yahoo Finance prices and share counts, SEC EDGAR XBRL 10-K facts (filed on or before the as-of date), the Ken French library (factors end **2026-08-31**), and the FRED 10-year yield.
- Notebooks `02` to `06` produce every figure. `results/summary.json` holds every number. A test checks each number on these slides against that file.
- `make all` rebuilds everything offline and runs the tests.

APPENDIX

Appendix A: rejected candidates

	Price	DCF value	vs price	Bear	Bull	Price-implied growth
COST	**$910.34**	**$410.96**	**-54.9%**	**$284.63**	**$496.94**	**38.5%**

V (first pick, archived). Dropped after M3: the base DCF was well below the price, even the bull case showed a loss, and the price implied far more FCF growth than V's history. See `results/archive/V_summary.json`.

COST (third on the shortlist). Every scenario, including bull at **$496.94**, is below the price. The price implies **38.5%** stage-1 growth.

All three were run with identical code and assumptions. Nothing was tuned to make a long case appear.

APPENDIX

Appendix B: assumptions

terminal growth	Assumption: roughly long-run nominal US GDP growth; kept below the 10y yield
fade years	Assumption: growth fades linearly from stage-1 to terminal over 10 years
scenario wacc shift	Assumption: +/-0.5pp WACC in bear/bull
trading cost bps	Assumption: 10 bp per one-way switch (commission plus spread for a mega-cap)
sma grid	Assumption: candidate trend-filter windows (trading days)
train years	Assumption: walk-forward training window
cash return	Assumption: cash earns 0 in the backtest (conservative for the strategy)
beta window years	Convention: 5y of daily returns for the CAPM beta